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RESEARCH ARTICLE

Automated Classification of User Exercise Poses in Virtual Reality Using Machine Learning-Based Human Pose Estimation

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ABSTRACT Human Pose Estimation (HPE) using machine learning presents significant potential for objectively analyzing user movements within Virtual Reality (VR) environments, particularly for applications involving guided physical exercises or interactive tasks. Analyzing 2D video footage from VR sessions for HPE offers a computationally efficient approach for tracking user movements. However, accuracy can be compromised by various factors, including occlusions from clothing or the VR headset itself. This study develops and evaluates a robust HPE system specifically designed to accurately classify predefined poses performed by users within such VR environments. The proposed system utilizes MediaPipe for corporal landmark extraction from user images (for training) and employs a Stacking Classifier ensemble, with XGBClassifier as the meta-learner, to classify six key exercise poses. Key results demonstrate an overall pose classification accuracy of 94% on a dedicated test set, with certain poses like 'Butterfly Hug' and 'Arm' achieving 100% accuracy. These findings highlight the system's potential to provide reliable, quantitative assessments of user adherence to prescribed movements in immersive interactive environments, offering valuable data for applications requiring objective analysis of physical performance in VR.

INDEX TERMS Human pose estimation (HPE), MediaPipe, stacking classifier, virtual reality (VR), video-based pose analysis.

I. INTRODUCTION

The burgeoning field of machine learning-based human pose estimation (HPE) for video analysis offers transformative potential for objectively assessing body movements across diverse applications, particularly in health and therapy. Recent studies highlight the increasing prevalence of using vision-based systems for therapeutic and rehabilitative purposes, such as real-time physiotherapy assessments, AI-based telerehabilitation, and the use of optimized machine learning models for stroke rehabilitation [1], [2], [3]. Despite these advances, current methods for evaluating adherence

to prescribed physical exercises often rely on subjective user reports or manual observation, which can be prone to bias or lack quantitative granularity. There is a clear need for objective, data-driven tools that can automatically assess how users perform prescribed movements, especially in interactive virtual reality (VR) settings. Such tools can provide valuable, quantifiable insights, enabling more effective feedback mechanisms or progress tracking for a wide range of applications.

To address this need, the primary objective of this study is to develop and rigorously evaluate an HPE system capable of automatically classifying a predefined set of six key exercise poses from video recordings of users participating in VR sessions. This paper details the development and

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evaluation of an HPE system tailored for such a context. Notably, the broader VR application, which provides the environment for these exercises, is part of an ongoing parallel collaboration with the Faculty of Medicine at King Mongkut's Institute of Technology Ladkrabang (KMUTL), Thailand. That collaboration focuses on creating and studying a VR application (developed with Unreal Engine 5.2 for the Meta Quest 2) specifically aimed at facilitating user engagement with prescribed exercises. A comprehensive discussion of the VR application's design and development is beyond the scope of this work. However, we detail the VR activities and the specific poses elicited in the Methodology section (Section III). These details include the poses detectable and non-detectable by our vision-based system, as they pertain to the HPE system requirements. The key contributions of this paper are the following: (1) We designed an HPE pipeline utilizing MediaPipe [4] for landmark extraction from still images (for model training) and a Stacking Classifier ensemble for pose classification. (2) This includes a systematic evaluation of different components and a comparison of MediaPipe against a representative YOLOv11 model. (3) We present a comprehensive analysis of the final model's classification accuracy on a dedicated test set. (4) Finally, we investigate the system's efficiency for video processing, focusing on the impact of frame skipping and resolution adjustments to assess its practicality for real-world application.

This paper is organized as follows: Section II discusses related works in the field of HPE and its applications. Section III explains the research methodology employed for developing the pose estimation system, including details of the VR exercise protocol and the HPE model architecture. Section IV details the experiments conducted to evaluate the system's performance and efficiency. Section V presents a discussion of our results, including a comparison with other studies and the limitations of our work. Finally, Section VI presents our conclusions and outlines future research directions.

II. LITERATURE REVIEW

VR is increasingly recognized as a transformative tool within the medical and wellness fields. It offers significant advancements for therapeutic interventions and patient rehabilitation by providing immersive and interactive environments [5], [6]. Specifically, VR is effectively utilized in contexts such as mental well-being programs where it can help individuals engage with guided exercises by creating controlled and safe spaces [7], [8]. Furthermore, in patient rehabilitation, VR provides engaging and interactive exercises that can enhance patient motivation and adherence to prescribed regimens. It is particularly beneficial in aiding the recovery of motor functions through repetitive, task-oriented training [9], [10]. The capacity of VR technology to support personalized exercise plans and potentially offer real-time feedback on movement or exercise performance is key to enhancing the efficacy of such programs [11].

The application of VR in these contexts often involves programs with specific physical postures and movements (such as those found in stress management or mindfulness protocols). This underscores a critical need for robust methods to objectively analyze user performance and adherence to these prescribed exercises. Such analysis can provide crucial feedback to both users and facilitators (e.g., therapists or trainers). This review, therefore, examines key advancements in HPE techniques, relevant datasets, and related machine learning strategies, contextualizing their applicability and limitations concerning the classification of defined exercise poses within VR interactive environments.

HPE has evolved into a vital tool for understanding human motion, with foundational 2D whole-body systems like OpenPose demonstrating the ability to localize multiple body keypoints from single images [12]. Despite such advancements, achieving consistently high accuracy in HPE remains challenging, primarily due to factors like occlusion. Occlusion occurs where body parts are obscured by clothing, environmental objects, self-occlusion, or sensing technology limitations. Addressing occlusion is paramount for reliable real-world HPE. Efforts in this area include specialized datasets like Dripe [13] for driving contexts and the Kanoya dataset used to retrain OpenPose for high-occlusion sports scenarios [14]. Complementing these are advanced algorithmic techniques. For instance, Xu and Wang proposed the Multi-Scale Structure-Aware Semantic GCN (MSS-GCN), which leverages geometric constraints and multi-scale structural relationships to compensate for missing data due to occluded keypoints [15]. Comprehensive frameworks like DAG tackle occlusion via targeted data augmentation, attention mechanisms, and graph-based models that explore body structure knowledge [16]. Architectures such as HRNeXt focus on capturing rich contextual information from high-resolution feature maps to enhance feature adaptability in occluded scenarios [17].

Beyond direct occlusion handling, research in HPE also focuses on domain adaptation and transfer learning to enhance model generalization, especially when applying models to new environments or specific tasks. Leveraging anatomical knowledge, for example, can constrain pose predictions to be anatomically plausible, thereby improving generalization [18]. Strategies for transferring knowledge from data-rich source domains to target domains with limited annotations are also crucial; Zhang et al. demonstrated that models trained on synthetic data can be effectively adapted to real-world scenarios [19]. These approaches are vital for developing robust HPE systems that can perform reliably in diverse settings, including specialized VR exercise environments.

The application of HPE extends to the detailed analysis of specific exercises and physical movements, particularly in rehabilitation and training. A core aspect is defining pose correctness, often by comparing user movements against ideal postures [20], with machine learning algorithms evaluating form based on labeled datasets of correct and

incorrect patterns [21]. A significant outcome is the ability to provide targeted, often real-time corrective feedback, including visual aids or reports on movement quality, even facilitating remote assessments [22]. The capacity to accurately classify foundational poses is a critical prerequisite for developing sophisticated assessment and feedback systems. This capacity, as addressed in our study, is vital for any guided exercise program within VR. Other relevant advancements include 3D HPE, exemplified by VIBE which combines CNNs and GRUs [23], and the application of real-time HPE frameworks like MediaPipe [4] for tasks such as sign language recognition [24] and yoga pose analysis [25], the latter emphasizing tailored model selection for specific poses.

Despite these diverse advancements, the specific domain remains an area requiring further exploration. This includes accurately and objectively assessing a range of prescribed physical exercises performed by users within general VR environments. Traditional evaluation methods for exercise adherence or form can be subjective or cumbersome. The unique VR context, while offering immersion, also introduces challenges for vision-based HPE, such as occlusions from the headset, controllers, or user interaction with their physical environment. Existing HPE methods, while addressing occlusion or enabling exercise analysis in other domains, may not be directly optimized for the specific set of exercises, interaction patterns, and data characteristics encountered when users perform guided movements within various VR applications.

Beyond foundational approaches, recent research has explored advanced applications of these techniques. For instance, the work by Anwar et al. [26] presented a comprehensive system for real-time human movement analysis, highlighting key advancements in tackling challenges like varying body sizes and lighting conditions. Similarly, the study by Anwar et al. [27] introduced a robust methodology for classifying complex human movements, emphasizing the importance of specific data preprocessing and an optimized model architecture to improve accuracy in real-world scenarios. Furthermore, recent studies demonstrate the increasing prevalence of machine learning-based pose analysis in therapeutic contexts. For instance, a system utilizing the YOLO Pose algorithm was found to be effective for real-time physiotherapy assessments, with a semantic framework translating raw pose data into clinically interpretable parameters like strength and balance [1]. Other research has successfully employed 3D pose estimation technology like Blazepose for AI-based telerehabilitation, showing it to be a viable and effective alternative for remote exercise interventions [2]. The application of advanced machine learning models, such as an optimized Bi-LSTM model, has also shown promise in stroke rehabilitation by providing personalized and efficient therapy plans [3]. These works collectively highlight the significant potential of applying sophisticated pose analysis to healthcare, underscoring the demand for objective, data-driven systems.

This study aims to address this specific technological gap by developing and evaluating an HPE system tailored for classifying a defined set of exercises performed within a VR setting. Our approach involves training a model on a dedicated dataset of these specific poses, leveraging robust landmark detection provided by MediaPipe, and employing a Stacking Classifier ensemble designed to effectively manage the high-dimensional data characteristic of pose estimation tasks. The goal is to provide a reliable method for objective movement analysis in VR, contributing to the broader field of interactive system assessment and potentially enabling more effective guided exercise experiences.

III. METHODOLOGY

This research aims to develop and evaluate an algorithmic system for detecting and assessing human pose during VR based stress reduction interventions. The methodology encompasses two primary components: the design and functionality of the VR application delivering the stress reduction activities, and the development of the pose assessment algorithm. The VR application provides guided exercises, while the algorithm is designed to automatically evaluate user adherence to these guided movements. The output of this pose assessment can subsequently be utilized to provide feedback on user performance.

A. VR APPLICATION DESIGN AND STRESS REDUCTION PROTOCOL

The VR application was designed in collaboration with the Faculty of Medicine at King Mongkut's Institute of Technology Ladkrabang (KMUTL) to guide users through a structured stress reduction protocol. The protocol comprises three distinct phases:

- 1) **Muscle Relaxation Phase:** Participants are guided to perform ten targeted progressive muscle relaxation exercises. Each exercise involves tensing a specific muscle group for approximately 5 seconds, followed by release. The targeted muscle groups include: calves, arms (biceps/triceps), fists, thighs (quadriceps/hamstrings), shoulders (shrugs/rolls), feet (curling/pointing), face (e.g., clenching jaw, furrowing brow), buttocks, abdomen, and back (e.g., arching).
- 2) **Butterfly Hug (BF_Hug) Phase:** Users engage in the Butterfly Hug technique, performing rhythmic, alternating arm movements across the chest for a duration of 5 minutes. Guidance is provided through visual and/or auditory cues within the VR environment.
- 3) **Breathing Meditation Phase:** Participants undertake guided breathing exercises for 3 minutes. Visual and/or auditory cues prompt specific durations for inhalation (4 seconds), breath hold (3 seconds), and exhalation (4 seconds).

This multi-component approach integrates established techniques for stress reduction. However, inherent limitations in current computer vision capabilities, particularly when



FIGURE 1. Visual examples of the six specific stress-reduction poses classified in the VR therapy assessment protocol.

users wear VR headsets and regular clothing, restrict the detectability of certain poses. For instance, facial muscle exercises (e.g., ‘Face’ pose involving clenching) are obscured by the VR headset, and abdominal clenching (‘Abdomen’ pose) may not be visually discernible through clothing. Consequently, from the set of activities, six poses were selected for automated video-based assessment due to their more distinct visual cues, thereby facilitating more reliable analysis by the computer vision system. These selected poses are detailed in Table 1. Examples of detectable and non-detectable poses are illustrated in Figure 1(a) and (b), respectively. To enable this video-based assessment, user activity during the VR application usage was recorded, with these recordings serving as the input data for the subsequent pose estimation and classification system.

TABLE 1. Qualitative detectability assessment of the six stress-reduction poses based on visual complexity and potential for self-occlusion in the VR environment.

Pose	Detectable	Pose	Detectable
BF_Hug	✓	Foot	×
Calves	✓	Face	×
Arm	✓	Buttocks	×
Fist	✓	Abdomen	×
Thighs	✓	Back	×
Shoulder	✓	Breathing	×

B. PROPOSED AUTOMATIC ASSESSMENT SYSTEM

The primary objective of this research is to develop a robust HPE system capable of automatically assessing user posture from video recordings captured during VR stress reduction exercises. The methodology for training this HPE system, as detailed in this section, utilized a curated dataset of still images representing key therapeutic poses. This model development process leveraged functionalities from OpenCV [28] and MediaPipe, adhering to a structured three-stage pipeline based on these still images: (1) data

acquisition and preparation from still images, (2) landmark extraction and data formatting, and (3) model training. An overview of this training pipeline for the HPE system is depicted in Figure 2. For the training phase, landmark detection is performed on each collected still image using MediaPipe’s Holistic model. This model yields 75 distinct corporal landmarks per image, each characterized by its semantic label (identifying the body part), 3D coordinates (x, y, z), and a visibility score (v). This comprehensive landmark dataset derived from still images subsequently serves as input for training a multi-class Stacking Classifier model. The final trained model is then serialized, intended for future deployment in the analysis of user activity from video recordings. The specifics of each stage in this still-image-based training process are elaborated in the ensuing subsections.

1) DATA ACQUISITION, PREPARATION, AND LANDMARK EXTRACTION

a: DATASET ACQUISITION

A dataset comprising 4,024 still images was compiled for this study. These images were directly captured from 10 participants, each performing six distinct poses associated with established stress management techniques, including targeted muscle tension-release exercises and rhythmic ‘Butterfly Hug’ movements. This targeted selection ensures the model’s relevance to the postures encountered in the VR application’s stress reduction protocol. The distribution of images across each pose class is detailed in Table 2.

b: IMAGE PREPROCESSING

To ensure data consistency for model training and to optimize for accurate landmark detection, all collected still images were uniformly processed to a final input resolution of 768×1024 pixels. This preprocessing step involved scaling each image to fit within these target dimensions while

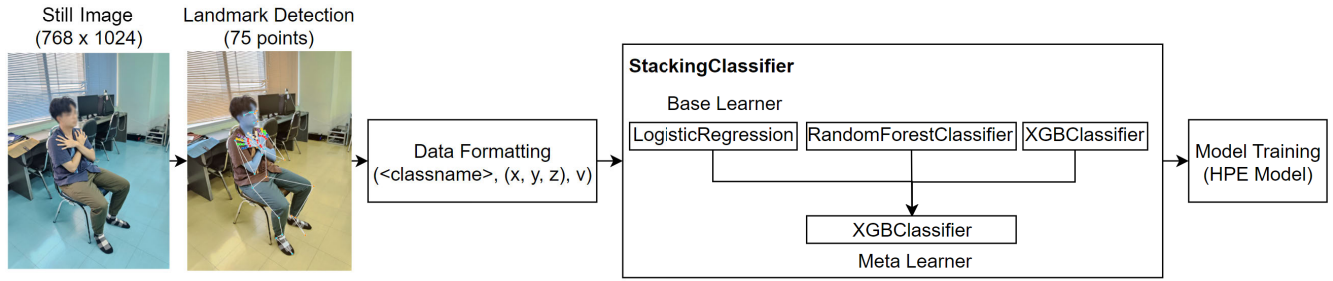


FIGURE 2. Overview of the training pipeline for the HPE classification system, detailing stages from still image input, landmark detection using MediaPipe’s Holistic model, and subsequent training of a multi-class Stacking Classifier.

TABLE 2. Distribution of the total dataset images across the six stress-related pose classes, indicating the number of samples used for training and validation.

Pose	No. of Images	Pose	No. of Images
BF_Hug	676	Calves	647
Arm	640	Fist	661
Thighs	609	Shoulder	791
Total Images: 4024			

maintaining its original aspect ratio. The resizing operations specifically utilized Bilinear interpolation, which is a common default method in libraries such as OpenCV [28], to ensure reasonable image quality during scaling. Any resulting empty areas within the 768×1024 frame, due to differences between the image’s scaled aspect ratio and the target frame’s aspect ratio, were then padded (e.g., with black pixels). This method ensured that the integrity of the depicted pose was preserved without introducing geometric distortion. The final 768×1024 resolution was selected as a balance between maintaining sufficient visual detail for effective pose estimation and managing computational load during model training and inference. All image manipulation operations, including resizing and padding, were performed using established image processing functionalities available in the OpenCV library [28].

c: LANDMARK DETECTION

The MediaPipe framework was employed for efficient and comprehensive landmark detection on the preprocessed still images. The model utilizes the human pose, and face landmark detectors, supplemented by the detailed hand detector extension, which in total identifies 75 corporal landmarks per image. These landmarks, illustrated for the ‘Butterfly Hug’ pose in Figure 3, encapsulate critical postural information, including joint angulation and body segment orientation, pertinent to stress state assessment.

2) DATA FORMATTING

To construct the feature set for model training, the following data elements were systematically extracted and recorded for each processed still image:

- **Pose Class Name:** A categorical label, y_i , identifying the specific pose depicted in image i . This serves as

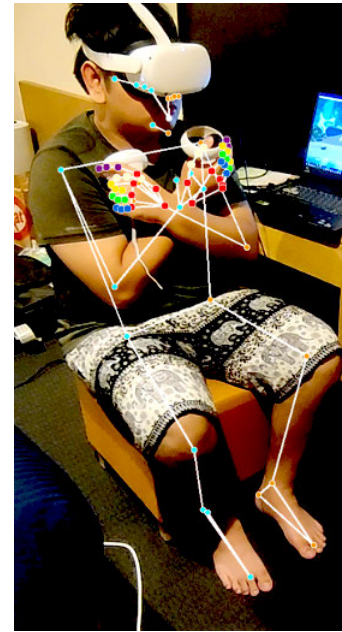


FIGURE 3. Illustration of the 75 Corporal landmarks detected by MediaPipe Holistic on a ‘Butterfly Hug’ pose, showing normalized coordinates used to encapsulate critical postural information for stress state assessment.

the ground truth for supervised model training, enabling the algorithm to learn associations between landmark configurations and specific prescribed exercises. For this study, $y_i \in \{C_1, C_2, \dots, C_K\}$, where $K = 6$ represents the total number of distinct pose classes.

- **Landmark Coordinates (x, y, z):** The 3D spatial coordinates for each of the $N_L = 75$ corporal landmarks detected by MediaPipe. For a given image i and landmark j (where $j \in \{1, \dots, N_L\}$), these are denoted as (x_{ij}, y_{ij}, z_{ij}) . These coordinates are normalized by MediaPipe: x_{ij} and y_{ij} are normalized to the range $[0.0, 1.0]$ by the image width and height respectively. The z_{ij} coordinate represents depth, with the depth at the midpoint of the hips being the origin, and its scale is approximately the same as that of the x_{ij} coordinate. This normalization provides a consistent representation of the user’s posture irrespective of image resolution or camera distance, which is crucial for the subsequent machine

learning model. However, no explicit post-processing normalization was applied to adjust for participant height or to further refine camera distance invariance beyond the default MediaPipe scaling.

- **Visibility Score (v):** A score, v_{ij} , provided by MediaPipe for each landmark j in image i . This score (typically ranging from 0.0 to 1.0) indicates the likelihood of that landmark being visible and unoccluded within the image.

From these extracted elements, a comprehensive feature vector, $\mathbf{x}_i$, was constructed for each image i . This vector concatenates the 3D coordinates and the visibility score for all N_L landmarks:

$$\mathbf{x}_i = [x_{i1}, y_{i1}, z_{i1}, v_{i1}, x_{i2}, y_{i2}, z_{i2}, v_{i2}, \dots, x_{iN_L}, y_{iN_L}, z_{iN_L}, v_{iN_L}]$$

Thus, each feature vector $\mathbf{x}_i$ resides in a D_{feat} -dimensional space, where $D_{feat} = 4 \times N_L = 4 \times 75 = 300$. The complete dataset D used for training and evaluation can then be represented as a set of N_S input-output pairs:

$$D = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N_S}$$

where $N_S = 4024$ is the total number of still images in our dataset.

This structured and comprehensive data formatting protocol ensures that the machine learning model is supplied with rich, quantitative information necessary to effectively learn the complex relationships between landmark geometries, their visibility, and the corresponding pose classes. This, in turn, is crucial for achieving accurate classification of the user's posture during the performance of guided exercises in the VR environment.

3) MODEL TRAINING

The classification task was framed as a multi-class classification problem. A Stacking Classifier ensemble was employed for its ability to combine multiple diverse base models and a meta-learner to leverage their strengths, thereby improving overall predictive performance and robustness. The workflow of the Stacking Classifier is illustrated in Figure 2.

Let M be the number of Level-0 base learners, denoted as $f_1, f_2, \dots, f_M$. For an input feature vector $\mathbf{x}_i$ (as defined in Section III-B2), each base learner f_m produces a set of predictions. To generate reliable inputs for the Level-1 meta-learner and to prevent data leakage, these Level-0 predictions are typically derived using a k -fold cross-validation procedure on the training dataset $D_{train} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N_{train}}$. The training dataset D_{train} is partitioned into k disjoint folds, $D_1, D_2, \dots, D_k$. For each fold D_j (where $j \in \{1, \dots, k\}$), the base learners f_m are trained on the remaining $k - 1$ folds ($D_{train} \setminus D_j$). Predictions, $\hat{\mathbf{p}}_{ij}^{(m)}$, are then made by these trained base learners for each sample $\mathbf{x}_i \in D_j$. These out-of-fold predictions (which could be class probabilities or direct class labels) are then collated. The set of these out-of-fold predictions from all M base learners for a sample $\mathbf{x}_i$ forms a

new feature vector, $\mathbf{x}'_i$:

$$\mathbf{x}'_i = [\hat{\mathbf{p}}_i^{(1)}, \hat{\mathbf{p}}_i^{(2)}, \dots, \hat{\mathbf{p}}_i^{(M)}]$$

A Level-1 meta-learner, denoted as F_{meta} , is then trained using these newly constructed feature vectors $\mathbf{x}'_i$ and the original ground truth labels y_i . For predicting on new, unseen data $\mathbf{x}_{new}$, the base learners f_m (now trained on the entire D_{train}) generate predictions, which are then fed as input to the trained meta-learner F_{meta} :

$$\hat{y}_{new} = F_{meta}([f_1(\mathbf{x}_{new}), f_2(\mathbf{x}_{new}), \dots, f_M(\mathbf{x}_{new})])$$

A critical design choice in this architecture is the selection of the meta-learner F_{meta} . We systematically evaluated three candidate algorithms for this role: Logistic Regression, Random Forest Classifier, and XGBClassifier. Each candidate classifier type first underwent hyperparameter optimization. This optimization was performed using RandomizedSearchCV, configured with 10 random search iterations over a pre-defined hyperparameter grid. The performance for each parameter combination was assessed using $k = 5$ -fold stratified cross-validation on the training dataset D_{train} . Stratification was employed to ensure that each fold maintained a class distribution representative of the overall dataset. The objective during this process was to maximize mean cross-validated accuracy. Parallel processing ($n_jobs = -1$) and a fixed random seed (1234) were employed for computational efficiency and reproducibility of the tuning process, respectively.

The efficacy of each optimized classifier type as a meta-learner was then compared based on its achieved mean cross-validated accuracy. The detailed comparative performance is presented in Section IV-A and Table 3. Based on this empirical evaluation, XGBClassifier was identified as the optimal meta-learner and was therefore selected for incorporation into the final Stacking Classifier model for this study. The base learners f_m selected for the final stacking ensemble were Logistic Regression, Random Forest, and an initial XGBClassifier.

Upon configuring the final Stacking Classifier with the chosen base learners and the optimized XGBClassifier as the meta-learner, the model was trained on the complete training dataset. Its generalization performance was subsequently assessed on a designated hold-out test set, the results of which are detailed in Section IV-C. Preliminary assessment during development indicated strong potential, as detailed in Section IV.

4) PERFORMANCE EVALUATION STRATEGY AND COMPARATIVE BASELINE

The generalization performance of the final proposed HPE system, configured as detailed in Section III-B3, was rigorously evaluated on a dedicated hold-out test set, D_{test} . This test set, comprising N_{test} samples, was kept separate from the training data D_{train} used for model development and hyperparameter optimization.

TABLE 3. Comparative classification accuracy (%) for various Meta-Learners used within the stacking classifier ensemble on the validation set.

Pose	XGBClassifier	LogisticRegression	RandomForest
BF_Hug	100	99	100
Calves	89	94	75
Arm	100	100	100
Fist	93	99	96
Thighs	83	64	81
Shoulder	97	64	96
Overall	94	92	91

The primary performance metric employed was classification accuracy. This is defined as the ratio of the number of correctly classified samples in the test set ($N_{correct_test}$) to the total number of samples in the test set (N_{test}):

$$\text{Accuracy} = \frac{N_{correct_test}}{N_{test}}$$

Further detailed insights into the classification performance across the K pose classes were obtained through the analysis of a confusion matrix, C . An element C_{jk} of this $K \times K$ matrix represents the number of test samples belonging to true class j that were predicted by the model as class k . From this matrix, the number of correctly classified samples can also be expressed as the sum of the diagonal elements (the trace of C):

$$N_{correct_test} = \sum_{j=1}^K C_{jj}$$

And the total number of test samples is the sum of all elements in C :

$$N_{test} = \sum_{j=1}^K \sum_{k=1}^K C_{jk}$$

Thus, accuracy can also be defined in terms of the confusion matrix elements:

$$\text{Accuracy} = \frac{\sum_{j=1}^K C_{jj}}{\sum_{j=1}^K \sum_{k=1}^K C_{jk}}$$

The detailed results of these evaluations are presented in Section IV.

To contextualize our system's performance, a comparative baseline was established using YOLOv11. This YOLOv11 model was selected as it represents a widely recognized, accessible, and robust general-purpose pose estimation framework, providing a strong benchmark for our specialized task of classifying defined exercises within a VR environment. The comparison with YOLOv11 aimed to validate the effectiveness of our chosen pipeline, particularly the utility of MediaPipe-derived landmarks and the Stacking Classifier architecture for this specific application.

While the literature review (Section II) discusses numerous advanced HPE techniques, a direct quantitative benchmarking against all such systems was determined to be beyond

the practical scope of this study. Such comprehensive comparisons are often confounded by variations in task focus (e.g., 3D shape estimation versus 2D pose classification), dataset specificity (with many systems evaluated on large public benchmarks that do not include our unique set of VR exercises), and the significant re-implementation or re-training efforts required for each external model on our custom dataset, especially when public code is not readily adaptable. Our evaluation strategy therefore prioritized a thorough assessment of our proposed system and a robust comparison against a relevant, accessible, and state-of-the-art general baseline to demonstrate its efficacy for the defined problem.

IV. EXPERIMENT AND RESULTS

This section presents the experimental evaluation of the proposed automated pose assessment system. The evaluation encompasses the selection of the pose estimation framework, the optimization of the classification model, the performance of the final system in classifying stress-reduction poses, and an analysis of system efficiency with respect to processing time and input resolution.

For comparative emphasis on YOLOv11's results against our methods, its system architecture in this context is depicted in Figure 4. It's important to note that YOLOv11 performs pose segmentation based on a more generalized, pose-wise approach, providing fewer detailed landmarks (in contrast to MediaPipe's 75 body indices, as discussed in Section III-B).

A. META-LEARNER SELECTION FOR STACKING CLASSIFIER

As outlined in the Methodology (Section III-B3), the selection of an optimal meta-learner for the Stacking Classifier involved the evaluation of three optimized candidates: XGBClassifier, Logistic Regression, and Random Forest Classifier. The comparative performance of these candidates, assessed by evaluating the overall pose prediction accuracy of each (using their respective optimized parameters as determined in Section III-B3) when applied to the training dataset through the 5-fold cross-validation procedure, is presented in Table 3.

As indicated in Table 3, XGBClassifier achieved the highest overall accuracy (94%), outperforming Logistic Regression (92%) and Random Forest Classifier (91%). Notably, XGBClassifier also demonstrated strong and consistent performance across most individual pose categories. Therefore, based on these empirical results, XGBClassifier was selected as the meta-learner for the final model configuration.

B. POSE ESTIMATION FRAMEWORK AND LANDMARK COMPARISON (MEDIPIPE VS. YOLOV11)

The initial phase of our experimental evaluation focused on selecting an optimal underlying framework for landmark detection, comparing MediaPipe Holistic with a representative YOLOv11 pose estimation model.

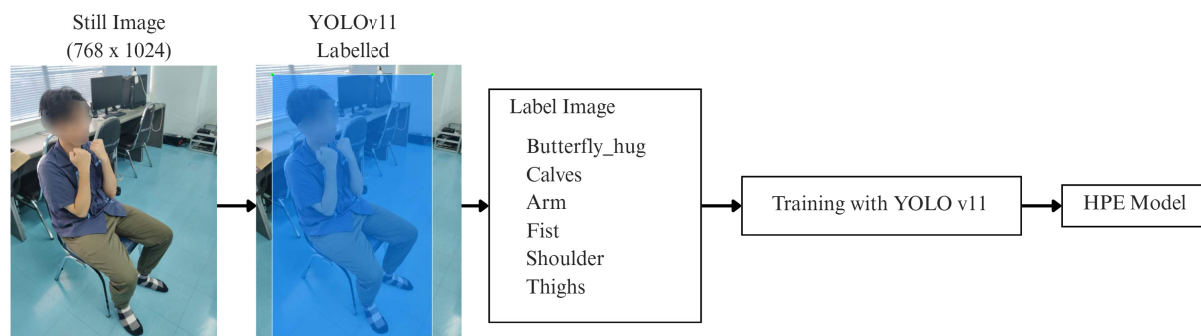


FIGURE 4. Overview of the alternative Human Pose Estimation (HPE) Pipeline utilizing YOLOv11 for comparative analysis, replacing MediaPipe as the core landmark detection module.

1) LANDMARK COVERAGE

A qualitative comparison of detectable corporal landmarks by each framework is presented in Table 4. MediaPipe demonstrated a more comprehensive landmark set, particularly

for distal limb segments such as hands (e.g., pinky, index, thumb) and feet (e.g., heel, foot index). These detailed landmarks are crucial for distinguishing subtle differences in the target stress-reduction poses, whereas the evaluated YOLOv11 model lacked several of these specific keypoints.

TABLE 4. Quantitative comparison of detectable corporal landmarks provided by MediaPipe Holistic (75 landmarks) versus the representative YOLOv11 framework, illustrating the feature extraction difference.

Landmark	MediaPipe	YOLOv11
Nose	Yes	Yes
Left eye inner	Yes	No
Left eye	Yes	Yes
Left eye outer	Yes	No
Right eye inner	Yes	No
Right eye	Yes	Yes
Right eye outer	Yes	No
Left ear	Yes	Yes
Right ear	Yes	Yes
Mouth left	Yes	No
Mouth right	Yes	No
Left shoulder	Yes	Yes
Right shoulder	Yes	Yes
Left elbow	Yes	Yes
Right elbow	Yes	Yes
Left wrist	Yes	Yes
Right wrist	Yes	Yes
Left pinky	Yes	No
Right pinky	Yes	No
Left index	Yes	No
Right index	Yes	No
Left thumb	Yes	No
Right thumb	Yes	No
Left hip	Yes	Yes
Right hip	Yes	Yes
Left knee	Yes	Yes
Right knee	Yes	Yes
Left ankle	Yes	Yes
Right ankle	Yes	Yes
Left heel	Yes	No
Right heel	Yes	No
Left foot index	Yes	No
Right foot index	Yes	No

2) POSE DETECTABILITY ASSESSMENT

The differences in landmark coverage directly influenced the theoretical capability of each framework to detect the full range of poses used in the VR application. Table 5 summarizes this qualitative assessment. While both frameworks could identify poses with gross bodily movements (e.g., ‘Butterfly Hug’, ‘Shoulder’), YOLOv11 was deemed less capable of providing sufficient detail for reliably distinguishing poses heavily reliant on hand and foot landmarks (e.g., ‘Calves’, ‘Arm’ (if involving fist clenching not just arm position), ‘Fist’, ‘Thighs’). The footnote indicates that a generic object tracking model based on YOLOv11 might detect the presence of a person in these poses, but the specific keypoints for fine-grained classification, as provided by MediaPipe, were often absent or insufficient with the

TABLE 5. Qualitative assessment of detectability based on landmark sufficiency: an analysis of whether the available landmarks from MediaPipe and YOLOv11 are sufficient to uniquely define the required six exercise poses.

Pose	MediaPipe	YOLOv11
Butterfly Hug	Yes	Yes
Calves	Yes	No*
Arm	Yes	No*
Fist	Yes	No*
Thighs	Yes	No*
Shoulder	Yes	Yes
Foot	No	No
Face	No	No
Buttocks	No	No
Abdomen	No	No
Back	No	No
Breathing	No	No

YOLOv11 may detect general movement but lacks specific landmarks for robust classification equivalent to MediaPipe for these poses. An object tracking variant might detect presence but not detailed pose.

evaluated YOLOv11 pose model. Several poses (e.g., ‘Foot’, ‘Face’, ‘Buttocks’, ‘Abdomen’, ‘Back’, ‘Breathing’) were considered non-detectable by either vision-based system due to occlusion by the VR headset, clothing, or the subtle nature of the muscular contraction, as initially outlined in the Methodology (Section III).

3) QUANTITATIVE CLASSIFICATION ACCURACY

To further compare the frameworks, both MediaPipe and YOLOv11 were used to classify the six target poses on a standardized test set of 100 images per pose class. The classification accuracies are presented in Table 6. The system employing MediaPipe-derived landmarks demonstrated markedly superior accuracy (e.g., 100% for ‘BF_Hug’ and ‘Arm’, 93% for ‘Fist’) compared to the YOLOv11-based approach, which struggled significantly with several poses (e.g., 24% for ‘Arm’, 0% for ‘Fist’ and ‘Thighs’). Consequently, MediaPipe Holistic was selected as the landmark extraction framework for the proposed assessment system. Note that the meta-learner in our proposed is selected based on the best perform criteria, detail will be explained in Section IV-A.

TABLE 6. Final pose classification accuracy comparison (%) between the Proposed MediaPipe-based Stacking Classifier and the benchmark YOLOv11-based System on the dedicated test set (100 images per pose).

Model	BF_Hug	Calves	Arm	Fist	Thighs	Shoulder
Mediapipe	100	89	100	93	83	97
YOLOv11	84	66	24	0	0	9

C. PERFORMANCE OF THE PROPOSED POSE CLASSIFICATION SYSTEM

The performance of the final proposed system, comprising MediaPipe Holistic for landmark extraction and a Stacking Classifier with an XGBClassifier meta-learner, was evaluated on 100 images per pose class. The confusion matrix in Table 7 details the classification performance for the six target poses.

TABLE 7. Confusion matrix (%) for the proposed MediaPipe-based stacking classifier on the test set, illustrating per-class true positives and misclassifications (Overall accuracy: 94%).

	BF_Hug	Calves	Arm	Fist	Thighs	Shoulder
BF_Hug	100	0	0	0	0	0
Calves	0	89	0	0	10	1
Arm	0	0	100	0	0	0
Fist	0	0	7	93	0	0
Thighs	0	17	0	0	83	0
Shoulder	0	1	0	1	1	97

Our system achieved an impressive accuracy of 100% for ‘BF_Hug’ and ‘Arm’ poses, with all test images for these classes being correctly classified. Similarly, ‘Shoulder’ pose classification yielded a high accuracy of 97%. The ‘Fist’ pose also demonstrated a high correct classification rate, with only minor misclassifications (7% attributed to ‘Arm’ class).

The lowest accuracies were observed for ‘Calves’ (89%) and ‘Thighs’ (83%). These two poses exhibited the most significant mutual misclassification, with 10% of ‘Calves’ instances mistaken for ‘Thighs’, and 17% of ‘Thighs’ instances mistaken for ‘Calves’. This confusion is primarily attributed to the subtle visual distinction between these poses from typical camera perspectives, where the primary differentiator (foot pointing direction) can be challenging to capture and interpret reliably. This detailed information from the confusion matrix allows for a deeper understanding of the model’s strengths and areas for potential improvement in classifying specific stress-related poses.

Further evaluation of the system’s performance utilized Precision, Recall, and F1-Score. The formulas for these metrics are:

$$Precision = \frac{N_{True_Positive}}{N_{True_Positive} + N_{False_Positive}}$$

$$Recall = \frac{N_{True_Positive}}{N_{True_Positive} + N_{False_Negative}}$$

$$F1 - Score = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

The best results for all three metrics were achieved by the ‘Butterfly Hug’ pose, with Precision, Recall, and F1-Score all equal to 1.0, indicating the highest level of accuracy. The next best was the ‘Shoulder’ pose, with an F1-Score of 0.98, reflecting a high average accuracy. Following these were the ‘Arm’ and ‘Fist’ poses, both with F1-Scores of 0.96, as shown in Table 8.

TABLE 8. Precision, Recall, and F1-Scores (%) derived from the confusion matrix for the proposed stacking Classifier, detailing the per-class performance metrics.

	Precision	Recall	F1-Score
BF_Hug	1	1	1
Calves	0.83	0.89	0.86
Arm	0.93	1	0.96
Fist	0.99	0.93	0.96
Thighs	0.88	0.83	0.85
Shoulder	0.99	0.97	0.98

D. SYSTEM EFFICIENCY EVALUATION FOR VIDEO PROCESSING

Given that the ultimate application of the trained model is intended for analyzing video recordings from the VR system, we evaluated processing efficiency under simulated application conditions.

1) IMPACT OF FRAME SKIPPING ON PROCESSING TIME

Given that the intended application of the developed HPE system involves analyzing video recordings of users engaged in VR stress reduction exercises—typically lasting 15-18 minutes—initial tests revealed that processing every frame of such videos was computationally intensive and

resulted in prohibitively long processing times. To address this challenge and enhance the system's practicality for near real-time feedback or efficient post-session analysis, a frame-skipping algorithm was developed and evaluated.

The efficacy of this technique was assessed by comparing the processing times for ten representative video clips (durations ranging from 14.50 to 16.51 minutes, captured at 1920×1080 resolution and 30 fps) with and without the frame-skipping algorithm implemented. The comparative results are detailed in Table 9.

TABLE 9. Comparison of total video clip processing times (minutes), demonstrating the efficiency gain achieved by implementing the dynamic frame skipping technique (Input: 1920×1080 @ 30 fps).

Clip ID	Duration (min)	No Skip (min)	Skip (min)	Saved (min)
1	16.17	56.00	6.37	49.23
2	15.31	57.14	4.30	52.44
3	14.50	55.54	4.37	51.17
4	15.32	58.38	3.58	54.40
5	16.51	68.17	4.38	63.39
6	15.27	55.07	3.34	51.33
7	15.32	56.31	3.46	52.45
8	16.01	59.09	4.38	54.31
9	15.16	56.31	3.39	52.52
10	15.32	58.57	4.42	54.13

All results were tested at a resolution of 1920×1080 @ 30 fps.

The Frame Skipping system dynamically adjusts video processing based on landmark detection and pose stability to optimize efficiency. The system operates according to the following procedure (visually detailed in Figure 5):

- 1) Initial Check: It first checks for the presence of all required body landmarks from the input image.
- 2) Handling Incomplete Data: If landmarks are incomplete for 5 consecutive frames, the system skips the next 30 frames, anticipating continued poor detection.
- 3) Handling Stable Poses: If landmarks are complete, the system predicts the current pose. To avoid redundant processing during stable periods, it skips 30 frames if the same pose is detected for 10 consecutive frames, continuing to skip until a pose change is identified.
- 4) Handling Unstable Changes: When a new pose is detected, the system checks for an unstable pose change over 5 consecutive frames and, if confirmed, skips 30 frames due to the unstable prediction.
- 5) Specific Termination Rule: A termination rule is applied for the 'Butterfly Hug' pose. If it persists for over 200 frames and is then followed by a new pose, the prediction process immediately stops, since any subsequent poses would fall outside the target exercise set, thus efficiently reducing unnecessary computation.

The introduction of frame skipping yielded a substantial reduction in processing duration. On average, the processing time per video clip was reduced from approximately 58.0 minutes to just 4.2 minutes, resulting in an average time saving of 52.8 minutes per clip. This constitutes an average processing speed-up of approximately 13.8 times

compared to analyzing all frames. Crucially, this significant enhancement in processing efficiency was achieved while maintaining the prediction performance of the pose classification model at a level deemed acceptable for the reliable interpretation of the overall sequence of therapeutic exercises. Validation during the development process confirmed that the chosen frame skipping strategy did not critically impair the system's ability to track user engagement with the prescribed poses throughout the session.

2) IMPACT OF INPUT RESOLUTION ON PROCESSING TIME

We further investigated whether reducing the input video resolution from Full HD (1080p) to 720p could yield additional processing time benefits when using the frame-skipping technique. The results, presented in Table 10, were inconclusive. Reducing resolution to 720p did not consistently decrease processing times; in several instances (Clips 4-10), 720p processing was comparable or slightly slower than 1080p. Furthermore, preliminary analyses during development suggested a potential degradation in landmark detection accuracy at 720p. Given these findings, the 1080p resolution was maintained as the standard input for the assessment system to ensure optimal accuracy and more predictable processing performance.

TABLE 10. Analysis of the trade-off between processing time and resolution: Comparison of total video clip processing times (minutes) using frame skipping at full high definition (1080p) versus High definition (720p) resolution.

Clip ID	Duration (min)	Time (1080p)	Time (720p)	Diff. (min)
1	16.17	6.37	5.30	1.07
2	15.31	4.30	4.02	0.28
3	14.50	4.37	4.12	0.25
4	15.32	3.58	4.03	-0.05
5	16.51	4.38	4.50	-0.12
6	15.27	3.34	3.45	-0.11
7	15.32	3.46	3.48	-0.02
8	16.01	4.38	5.12	-0.34
9	15.16	3.39	3.40	-0.01
10	15.32	4.42	4.44	-0.02

E. DETAILED PER-INSTANCE PERFORMANCE ANALYSIS

To gain more granular insights into the system's performance beyond aggregate metrics, a detailed analysis was conducted across ten distinct test instances (e.g., representing individual participant sessions or video clips, as used in the efficiency evaluations). Table 11 presents a breakdown of performance for each of the six classified poses across these instances, detailing the number of frames detected, the total summed confidence score for those frames, and the resulting average confidence. A qualitative 'Result' is also provided, categorizing system performance based on the average confidence scores achieved for each pose instance. Three distinct categories are used:

- 'Great' indicates an average confidence score ≥ 0.8 .

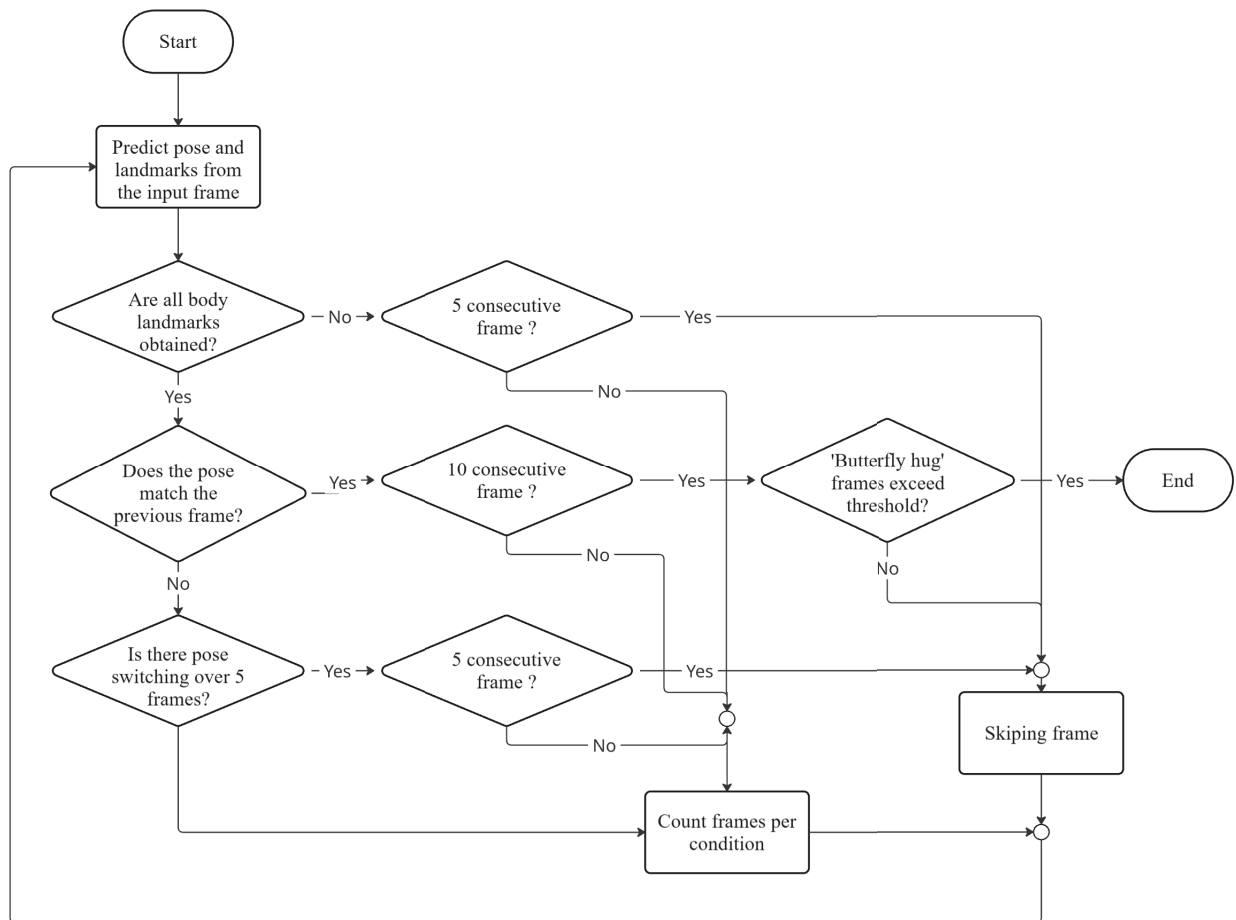


FIGURE 5. Workflow diagram detailing the dynamic frame skipping system, which is implemented to optimize video processing time by adaptively analyzing frame-to-frame changes before executing the full HPE classification.

- ‘Good’ indicates an average confidence score that is ≥ 0.3 but < 0.8 .
- ‘OK’ indicates an average confidence score < 0.3 .

These categories were established to provide a general overview of performance, acknowledging the application’s context where encouraging user engagement throughout the VR exercise sessions was a key consideration, rather than imposing strictly rigorous performance benchmarks.

The Frame Detection data in Table 11 reveals considerable variability in the duration or frequency for which different poses were detected across the instances. For example, the ‘BF_Hug’ pose was consistently detected for a substantial number of frames (ranging from 202 to 362 frames). ‘Fist’ also showed high frame detection in several instances (e.g., 513 frames in Instance 1, 589 in Instance 5), though it was detected for fewer frames in others (e.g., 37 in Instance 6, 38 in Instance 7). In contrast, poses such as ‘Thighs’ (e.g., as few as 10-11 frames in Instances 3 and 6) and ‘Calves’ (e.g., 6 frames in Instance 5, though 317 in Instance 1) exhibited greater variation in detection counts, potentially reflecting shorter or less frequent performance of these specific exercises by participants, or intermittent detection challenges.

Analysis of the “Average Confidence” scores provides crucial insights into the classifier’s certainty for each pose per instance. The ‘BF_Hug’ pose consistently demonstrated high average confidence, frequently at or above 0.96 (e.g., Instances 1, 3, 4, 5, 7, 8, 9). The ‘Arm’ and ‘Shoulder’ poses also generally yielded high average confidence scores across most instances (often >0.95 for ‘Arm’ and >0.92 for ‘Shoulder’ in several instances like 4, 7, 8, 9, 10 for ‘Arm’, and 4, 6, 7, 8, 9 for ‘Shoulder’).

Consistent with the aggregate confusion matrix findings (Section IV-C), which highlighted challenges in differentiating ‘Calves’ and ‘Thighs’, these poses exhibited greater variability and sometimes lower average confidence scores at the per-instance level. For ‘Calves’, Instance 5 showed a notably low average confidence of 0.351 (with only 6 frames detected), which would be categorized as ‘Good’ based on the defined thresholds. Instance 3 was also relatively low at 0.558 (‘Good’). However, other instances demonstrated moderate to good confidence for ‘Calves’ (e.g., Instance 7 at 0.665 (‘Good’), Instance 2 at 0.775 (‘Good’), and Instance 10 at 0.839 (‘Great’)), underscoring its variable performance. Similarly, ‘Thighs’ showed a range of average confidences, from a low of 0.422 in Instance 2 (‘Good’) to

scores above 0.8 in several other instances (e.g., 0.847 in Instance 3 ('Great'), 0.850 in Instance 5 ('Great'), 0.808 in Instance 7 ('Great')). This per-instance data reinforces that these subtly distinguished poses present a greater challenge to the classifier, leading to lower or more varied confidence in specific occurrences. The 'Fist' pose also showed some variability, with average confidence ranging from around 0.70 (Instance 3 and 9, categorized as 'Good') to over 0.98 (Instance 8, categorized as 'Great').

The qualitative "Result" column in Table 11, when populated according to these defined thresholds, would reflect these quantitative trends. Poses like 'BF_Hug', 'Arm', and 'Shoulder' would frequently receive 'Great' assessments. Conversely, 'Calves' and 'Thighs' would show a mix of 'Good' and 'Great', and potentially 'OK' if any instances fell below the 0.3 average confidence threshold, aligning with their observed variability.

This detailed per-instance analysis complements the overall system performance metrics by highlighting the consistency of pose detection and classification confidence across different scenarios or participants. It helps pinpoint specific poses or individual instances where the model may be less certain, providing valuable diagnostic information for potential future refinements of the system or for guiding the interpretation of its output in practical applications involving guided exercises.

V. DISCUSSION

The automated HPE system developed in this study, which achieved a robust 94% classification accuracy for stress-reduction poses, demonstrates the significant potential of applying computer vision to objective feedback in VR digital therapies. Our results are consistent with a broader trend in the literature that uses vision-based analysis for health applications, as highlighted by our comparative analysis.

Our findings align with studies that found algorithms like YOLO Pose to be effective for real-time physiotherapy assessments [1]. Similarly, our conclusion that AI-based systems are viable alternatives for a variety of health applications is supported by research using 3D pose estimation (Blazepose) in telerehabilitation [2] and optimized machine learning models for stroke rehabilitation [3]. This shows that our work is a part of a larger, interdisciplinary effort to leverage technology for improved health outcomes.

The robust 94% accuracy of our HPE system offers an objective, automated assessment tool for VR digital therapies, carrying direct clinical relevance. The immediate use case is providing precise, real-time feedback within the VR stress-reduction protocol to ensure user adherence. Future applications include remote stress monitoring, quantifiable movement reports for physiotherapy, and metrics for automated assessment in telerehabilitation platforms.

However, our system's performance metrics also reveal some key limitations. While we achieved high accuracy for distinct poses like the "Butterfly Hug," the challenges with

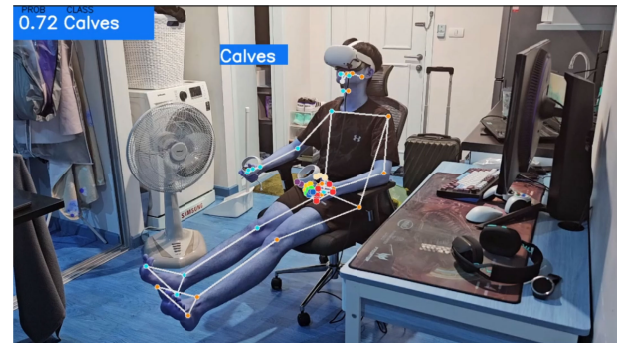
TABLE 11. Per-instance performance metrics, providing frame detection counts and confidence scores (Total/Average) for the ten video clips, used to qualitatively assess the system's reliability during continuous pose estimation.

Row	BF_Hug	Calves	Arm	Fist	Thighs	Shoulder
Frame Detection (frames)						
1	291	317	349	513	70	120
2	202	103	527	409	230	97
3	362	182	122	289	10	396
4	317	46	39	325	36	218
5	313	6	45	589	60	302
6	330	45	50	37	11	366
7	309	51	41	38	48	445
8	307	11	41	251	70	292
9	328	61	22	81	18	451
10	327	298	332	415	43	108
Total Confidence (score)						
1	289.488	229.097	225.004	460.627	36.703	63.125
2	154.468	79.787	386.326	355.655	97.056	77.415
3	349.079	101.561	94.913	201.677	8.471	303.896
4	309.186	29.692	38.980	310.780	30.076	201.390
5	308.540	2.104	44.574	545.268	51.006	244.625
6	279.266	36.489	44.293	36.054	7.839	353.261
7	300.366	33.92	40.203	32.284	38.807	428.935
8	301.311	8.729	40.054	247.474	60.274	269.209
9	321.717	50.284	21.783	56.785	15.108	427.007
10	308.431	250.126	322.852	397.579	30.260	92.754
Average Confidence						
1	0.995	0.723	0.645	0.898	0.524	0.526
2	0.765	0.775	0.733	0.870	0.422	0.798
3	0.964	0.558	0.778	0.698	0.847	0.767
4	0.975	0.645	0.999	0.956	0.835	0.924
5	0.986	0.351	0.991	0.926	0.850	0.810
6	0.846	0.811	0.886	0.974	0.713	0.965
7	0.972	0.665	0.981	0.850	0.808	0.964
8	0.981	0.794	0.977	0.986	0.861	0.922
9	0.981	0.824	0.990	0.701	0.839	0.947
10	0.943	0.839	0.972	0.958	0.704	0.859
Result						
1	Great	Good	Good	Great	Good	Good
2	Good	Good	Good	Great	Good	Good
3	Great	Good	Good	Good	Great	Good
4	Great	Good	Great	Great	Great	Great
5	Great	Good	Great	Great	Great	Great
6	Great	Great	Great	Great	Good	Great
7	Great	Good	Great	Great	Great	Great
8	Great	Good	Great	Great	Great	Great
9	Great	Great	Great	Good	Great	Great
10	Great	Great	Great	Great	Good	Great

"Calves" (89%) and "Thighs" (83%) indicate that our model struggles with more subtle movements and visual cues. Visual examples of these misclassification patterns are illustrated in Figure 6. This highlights a limitation also present in other studies. The lack of temporal analysis in our current system is a key limitation, as the dynamics of movement are often crucial for accurate pose assessment. Additionally, the study's dataset lacks diversity in terms of body types and



(a) Calves (Actual) misclassified as Thighs (Predicted)



(b) Thighs (Actual) misclassified as Calves (Predicted)

FIGURE 6. Visual examples of misclassification patterns between the “Calves” and “Thighs” Poses. (a) A user performing the “Calves” pose incorrectly classified as the “Thighs” pose. (b) A user performing the “Thighs” pose incorrectly classified as the “Calves” pose. These examples highlight the visual ambiguity and subtle differences that challenge the model’s differentiation between these two lower-body exercises.

backgrounds, which may limit the system’s generalizability in real-world settings.

VI. CONCLUSION AND FUTURE WORK

This research successfully developed and rigorously evaluated an automated HPE system for classifying six stress-reduction exercises within a VR therapeutic environment. Integrating MediaPipe Holistic for landmark extraction and employing a Stacking Classifier ensemble with an XGB-Classifier meta-learner, the system achieved a robust 94% overall classification accuracy. Notably, a frame-skipping technique yielded a significant 13.8-fold processing time reduction, crucial for practical application. This work’s primary contribution is the systematic development of an accurate and efficient HPE system tailored for novel VR stress-reduction, demonstrating the potential of computer vision for objective feedback in digital therapies.

Building on this study’s findings, future work will focus on several key areas to enhance the HPE system for VR stress reduction. We aim to improve pose discrimination and dynamics by incorporating temporal information and advanced neural network architectures (e.g., LSTMs, TCNs, GCNs). A critical next step is developing and integrating a real-time feedback version within the VR application, followed by user studies to evaluate its impact on engagement and stress reduction outcomes. We also plan to extend the system beyond mere classification to qualitative pose assessment, analyzing aspects like pose duration, stability, and subtle muscular tension. For broader impact, clinical validation and diverse user studies are essential, collaborating with practitioners to assess usability and correlation with established physiological measures. Furthermore, we’ll explore multimodal sensing (e.g., EMG, depth sensors) for visually challenging poses and investigate methods for personalization and adaptive learning based on individual user biomechanics. Finally, enhancing robustness to real-world variability and conducting longitudinal studies will be crucial to understand long-term system utility and sustained engagement.

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